MSG No. 127.397 P_C4/mbm [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4'/mm'm

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,6,7,8,9,14,15,16,18,19,20,21,26,27,28,29]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,4,5,10,11,12,13,17,22,23,24,25,30,31,32]

Table 2: Wyckoff site: 2b, site symmetry: 4'/mm'm

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,6,7,8,9,14,15,16,18,19,20,21,26,27,28,29]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,4,5,10,11,12,13,17,22,23,24,25,30,31,32]

Table 3: Wyckoff site: 2c, site symmetry: 4/mm'm'

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,6,9,10,11,14,20,21,23,24,28,29,31,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[4,5,7,8,12,13,15,16,17,18,19,22,25,26,27,30]

Table 4: Wyckoff site: 2d, site symmetry: 4/mm'm'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,6,9,10,11,14,20,21,23,24,28,29,31,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,5,7,8,12,13,15,16,17,18,19,22,25,26,27,30]

Table 5: Wyckoff site: 4e, site symmetry: m.mm'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[1,8,14,15,22,23,25,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,4,11,13,19,21,26,28]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[3,5,10,12,18,20,27,29]
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[6,7,9,16,17,24,30,31]

Table 6: Wyckoff site: 4f, site symmetry: m.mm'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[1,8,14,15,22,23,25,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,4,11,13,19,21,26,28]
3	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[3,5,10,12,18,20,27,29]
4	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[6,7,9,16,17,24,30,31]

Table 7: Wyckoff site: `4g`, site symmetry: `4'm'm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,15,16,18,19,28,29]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,12,13,17,22,31,32]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[4,5,10,11,23,24,25,30]`
4	$\left[0,\ \frac{1}{2},\ -z\right]$	`[7,8,9,14,20,21,26,27]`

Table 8: Wyckoff site: `4h`, site symmetry: `4m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6,28,29,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,7,8,25,26,27,30]`
3	$[0,\ 0,\ -z]$	`[9,10,11,14,20,21,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[12,13,15,16,17,18,19,22]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[1,15,22,32]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[2,13,19,28]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[3,12,18,29]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[4,11,21,26]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[5,10,20,27]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[6,16,17,31]`
7	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[7,9,24,30]`
8	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[8,14,23,25]`

Table 10: Wyckoff site: `8j`, site symmetry: `mm'2'.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,14,21,28]`
2	$[-y,\ 0,\ 0]$	`[2,11,24,31]`
3	$[y,\ 0,\ 0]$	`[3,10,23,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[4,13,22,25]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[5,12,17,30]`
6	$[0,\ -y,\ 0]$	`[6,9,20,29]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[7,16,19,26]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	`[8,15,18,27]`

Table 11: Wyckoff site: `8k`, site symmetry: `mm'2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,14,21,28]`
2	$\left[-y,\ 0,\ \frac{1}{2}\right]$	`[2,11,24,31]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\, 0,\, \frac{1}{2}\right]$	[3,10,23,32]
4	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[4,13,22,25]
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[5,12,17,30]
6	$\left[0,\, -y,\, \frac{1}{2}\right]$	[6,9,20,29]
7	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	[7,16,19,26]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}\right]$	[8,15,18,27]

Table 12: Wyckoff site: 8l, site symmetry: m.2m

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	[1,7,14,16]
2	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	[2,5,11,12]
3	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	[3,4,10,13]
4	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\, x,\, 0\right]$	[17,23,30,32]
6	$\left[-x,\, x+\frac{1}{2},\, 0\right]$	[18,21,27,28]
7	$\left[x,\, \frac{1}{2}-x,\, 0\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\, -x,\, 0\right]$	[22,24,25,31]

Table 13: Wyckoff site: 8m, site symmetry: m.2m

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[1,7,14,16]
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	[2,5,11,12]
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,4,10,13]
4	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[17,23,30,32]
6	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[18,21,27,28]
7	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	[22,24,25,31]

Table 14: Wyckoff site: 8n, site symmetry: m.2'm'

No.	position	mapping
1	$[x,\, x,\, 0]$	[1,14,23,32]
2	$[-x,\, x,\, 0]$	[2,11,21,28]
3	$[x,\, -x,\, 0]$	[3,10,20,29]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[4,13,19,26]
5	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, 0\right]$	[5,12,18,27]
6	$[-x,\, -x,\, 0]$	[6,9,24,31]
7	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[7,16,17,30]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, 0\right]$	[8,15,22,25]

Table 15: Wyckoff site: `8o`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,14,23,32]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,11,21,28]`
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,10,20,29]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[4,13,19,26]`
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,12,18,27]`
6	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[6,9,24,31]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16,17,30]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15,22,25]`

Table 16: Wyckoff site: `16p`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$[-y,\ x,\ 0]$	`[2,11]`
3	$[y,\ -x,\ 0]$	`[3,10]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	`[5,12]`
6	$[-x,\ -y,\ 0]$	`[6,9]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	`[8,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ 0\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[19,26]`
12	$[x,\ -y,\ 0]$	`[20,29]`
13	$[-x,\ y,\ 0]$	`[21,28]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	`[22,25]`
15	$[y,\ x,\ 0]$	`[23,32]`
16	$[-y,\ -x,\ 0]$	`[24,31]`

Table 17: Wyckoff site: `16q`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,14]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,11]`
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[3,10]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,12]`
6	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[6,9]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[19,26]`
12	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[20,29]`

continued ...

Table 17

No.	position	mapping
13	$\left[-x,\ y,\ \frac{1}{2}\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[22,25]
15	$\left[y,\ x,\ \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[24,31]

Table 18: Wyckoff site: 16r, site symmetry: .m'.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,28]
2	$[-y,\ 0,\ z]$	[2,31]
3	$[y,\ 0,\ z]$	[3,32]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[4,25]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[5,30]
6	$[0,\ -y,\ z]$	[6,29]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[7,26]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -z\right]$	[8,27]
9	$[0,\ -y,\ -z]$	[9,20]
10	$[y,\ 0,\ -z]$	[10,23]
11	$[-y,\ 0,\ -z]$	[11,24]
12	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[12,17]
13	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[13,22]
14	$[0,\ y,\ -z]$	[14,21]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z\right]$	[15,18]
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z\right]$	[16,19]

Table 19: Wyckoff site: 16s, site symmetry: ..m

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,16]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,12]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,13]
4	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[4,10]
5	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[5,11]
6	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[6,15]
7	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[7,14]
8	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[8,9]
9	$\left[x+\frac{1}{2},\ x,\ z\right]$	[17,32]
10	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[18,28]
11	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[19,29]
12	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	[20,26]
13	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	[21,27]
14	$\left[\frac{1}{2}-x,\ -x,\ z\right]$	[22,31]
15	$\left[x+\frac{1}{2},\ x,\ -z\right]$	[23,30]
16	$\left[\frac{1}{2}-x,\ -x,\ -z\right]$	[24,25]

Table 20: Wyckoff site: `16t`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,32]`
2	$[-x,\ x,\ z]$	`[2,28]`
3	$[x,\ -x,\ z]$	`[3,29]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	`[4,26]`
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	`[5,27]`
6	$[-x,\ -x,\ z]$	`[6,31]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[7,30]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -z\right]$	`[8,25]`
9	$[-x,\ -x,\ -z]$	`[9,24]`
10	$[x,\ -x,\ -z]$	`[10,20]`
11	$[-x,\ x,\ -z]$	`[11,21]`
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	`[12,18]`
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[13,19]`
14	$[x,\ x,\ -z]$	`[14,23]`
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[15,22]`
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16,17]`

Table 21: Wyckoff site: `32u`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x,\ z]$	`[2]`
3	$[y,\ -x,\ z]$	`[3]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$[-x,\ -y,\ z]$	`[6]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[7]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$[y,\ -x,\ -z]$	`[10]`
11	$[-y,\ x,\ -z]$	`[11]`
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[12]`
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[13]`
14	$[x,\ y,\ -z]$	`[14]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	`[15]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16]`
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[17]`
18	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	`[18]`
19	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[19]`
20	$[x,\ -y,\ -z]$	`[20]`
21	$[-x,\ y,\ -z]$	`[21]`
22	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[22]`
23	$[y,\ x,\ -z]$	`[23]`
24	$[-y,\ -x,\ -z]$	`[24]`
25	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	`[25]`

continued ...

Table 21

No.	position	mapping
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[27]
28	$[-x,\ y,\ z]$	[28]
29	$[x,\ -y,\ z]$	[29]
30	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[30]
31	$[-y,\ -x,\ z]$	[31]
32	$[y,\ x,\ z]$	[32]